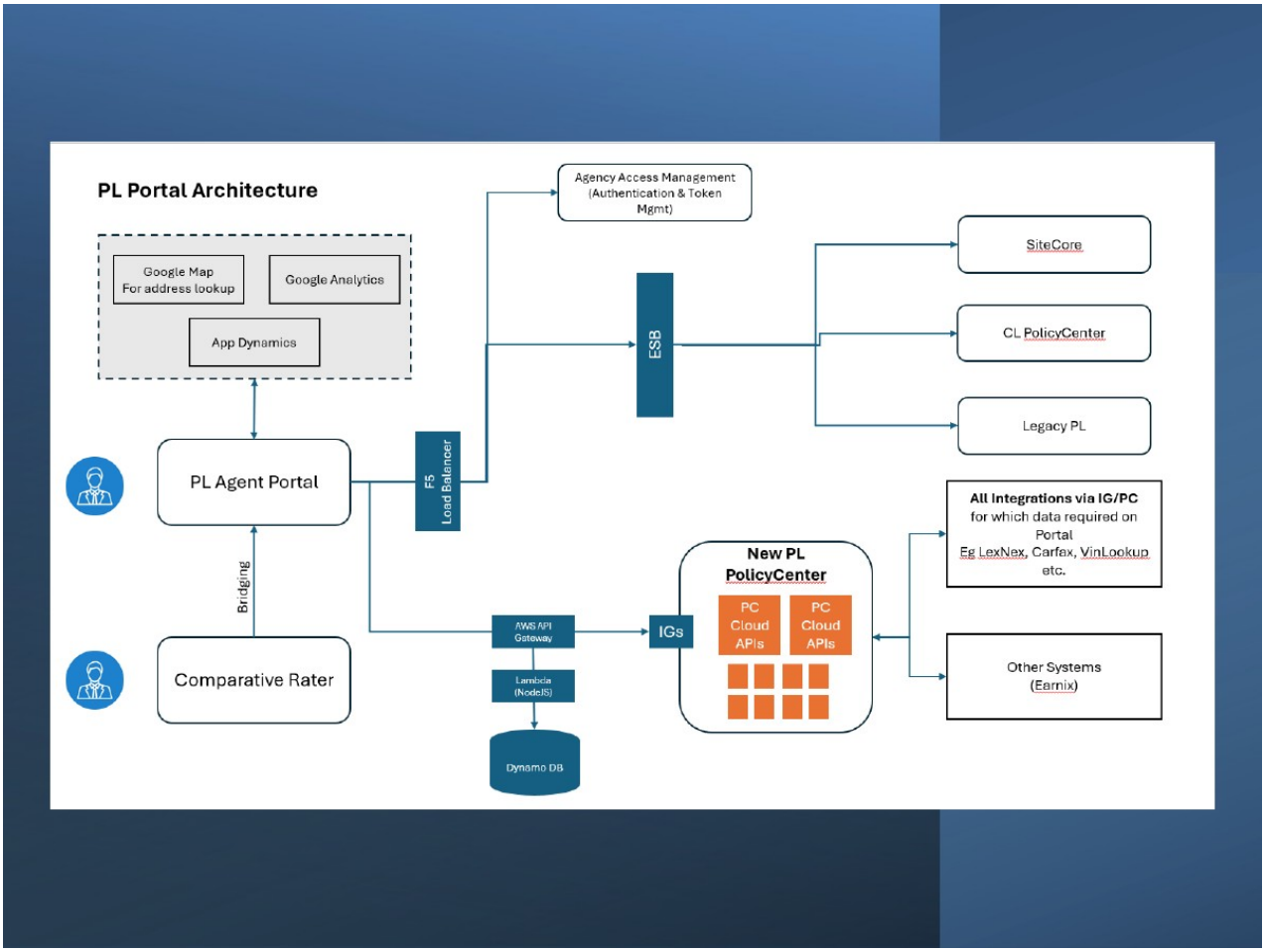
GRANGE PERSONAL LINES

Agent Portal 2.0 – Solution Architecture Document

End-to-End Architecture, Technical Governance & Delivery View

Prepared from the supplied Grange KYP presentation, PL Core4 technical architecture deck, project-information workbook, and UI screens.



Confidential – project architecture working document

1. Document Control

Item	Value
Client	Grange Personal Lines Insurance
Solution	Personal Lines Agent Portal 2.0 / PL Agent Portal
Primary role	Solution / Technical Architecture
Source material	KYP Agent Portal Presentation; PL Core4 Technical Architecture Approach; Project Information workbook; supplied portal screens
Architecture status	The supplied technical deck states the architecture was under discussion and subject to Architecture Board approval at that point in time.
Delivery approach	Agile/Scrum with Git-based development and CI/CD

Source note: the project workbook describes the platform as React.js with microfrontend architecture and mentions Guidewire Claim Center, while the supplied portal architecture decks describe integrations including PolicyCenter, CL PolicyCenter and Legacy PL. This document preserves those source statements rather than silently reconciling them.

2. Executive Summary

The Grange Personal Lines Agent Portal 2.0 is positioned as a unified agent-facing portal for policy, quote and transaction capabilities across Personal Lines, Commercial Lines and legacy environments. The KYP material identifies the objective as a unified portal covering Auto, Home, Umbrella, Legacy and CL systems, with integration through PolicyCenter, APIs and external systems.

The portal experience shown in the supplied screens includes a combined dashboard, global search, account/policy/quote views, quote initiation and detailed Auto/Home/Umbrella quote journeys, coverage selection, billing and payment. The architecture material emphasizes a modern React front end, API integration, cloud components, authentication/token management, observability and resilience.

Architecture outcomes

- Unified agent experience across multiple policy and legacy systems.
- Fast user experience with a stated target of response time within 2 seconds, subject to the response-time agreement with Policy.
- Scalable and secure portal architecture with reusable UI components.
- Integration isolation so external-system failures do not bring down the portal.
- Accessibility alignment to WCAG 2.1 AA.
- Engineering quality through Sonar quality gates and minimum 80% unit-test coverage.

3. Business & Functional Scope

Area	Capabilities evidenced in source	Approx. screen scope
Dashboard	Combined dashboard; open quotes; open policy changes; pending non-pay cancellations; recently viewed Accounts/Quotes/Policies; To-Dos; Customer Care; chat/feedback	3–4
Start a Quote	Initiate quote for Legacy PL, Commercial Lines and TrueCover PL	2
TrueCover Auto	Quote → purchase → issue	18–20
TrueCover Home	HO3, HO4, HO6 quote → purchase → issue	18–20
TrueCover Umbrella	Quote → purchase → issue	10–12
Account	Account list, filters, account details	2
Quote	Quote list, filters, quote details	2
Policy	Policy list, individual policy view	2
Transactions	Transaction list and detail	2
Policy services	Amend, documents, billing, claims, cancel, reinstate, rewrite	Varies

The supplied UI screens visually demonstrate the dashboard, product selection, account details, quote list, policy information, coverage selection and billing/payment stages.

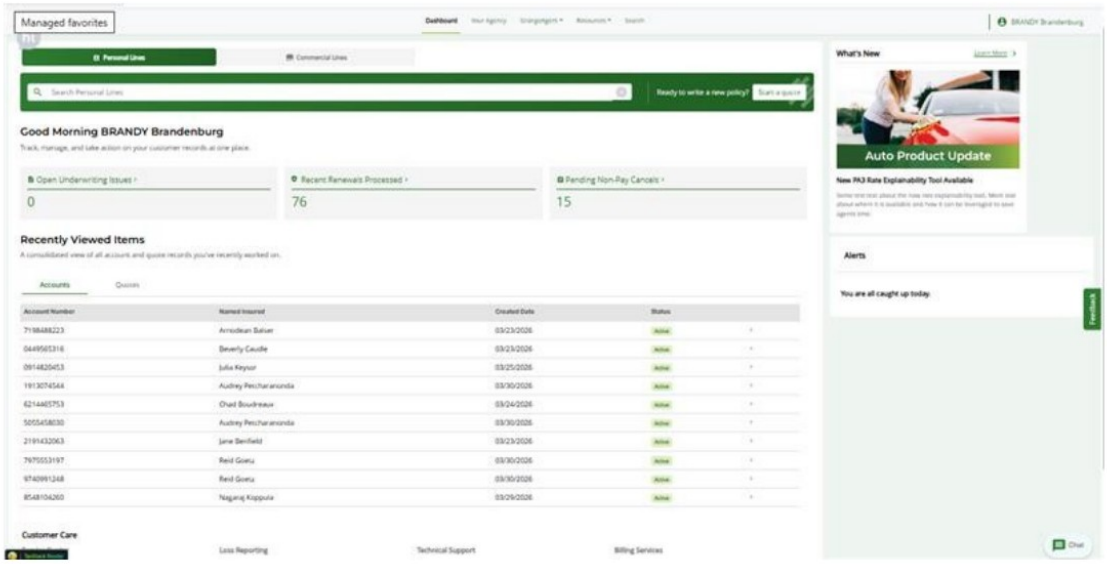


Figure: supplied portal screen – KYP presentation page 16.

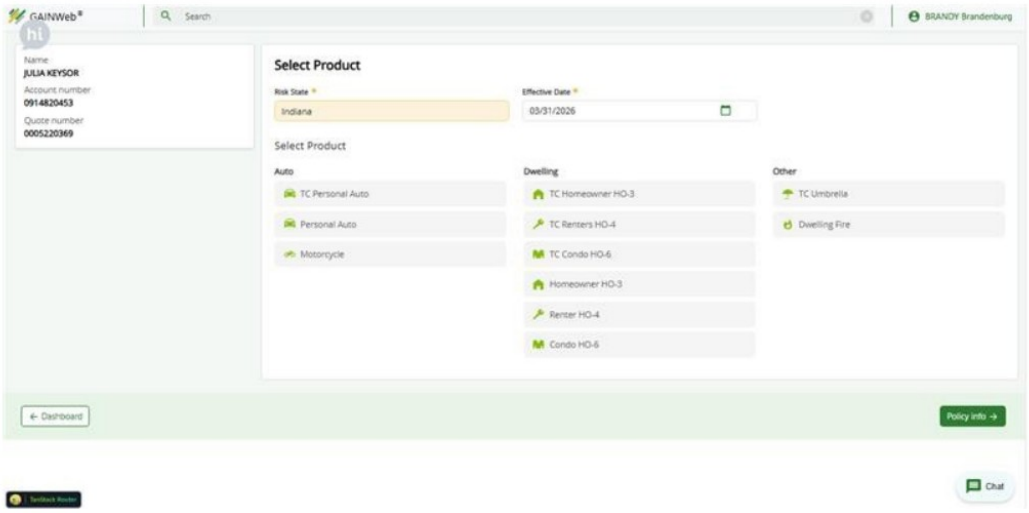


Figure: supplied portal screen – KYP presentation page 18.

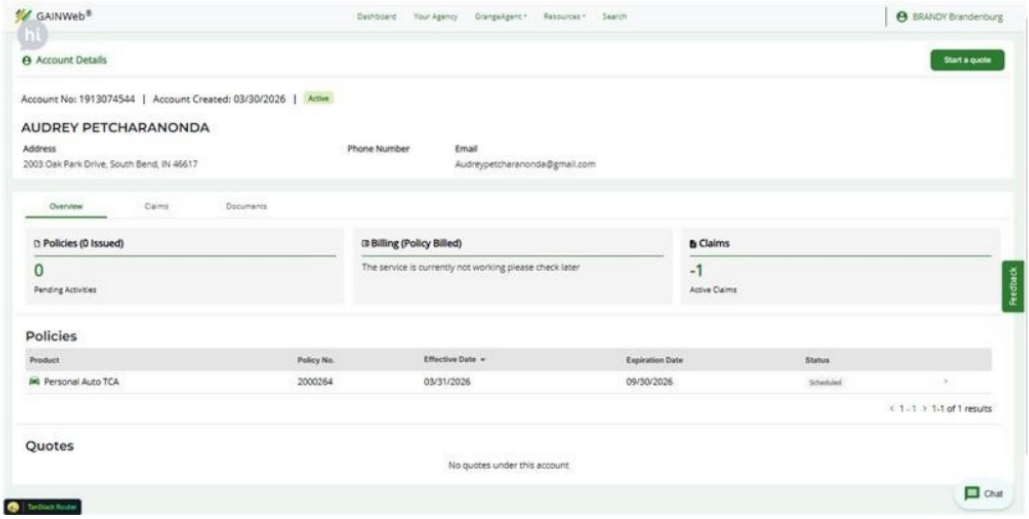


Figure: supplied portal screen – KYP presentation page 19.

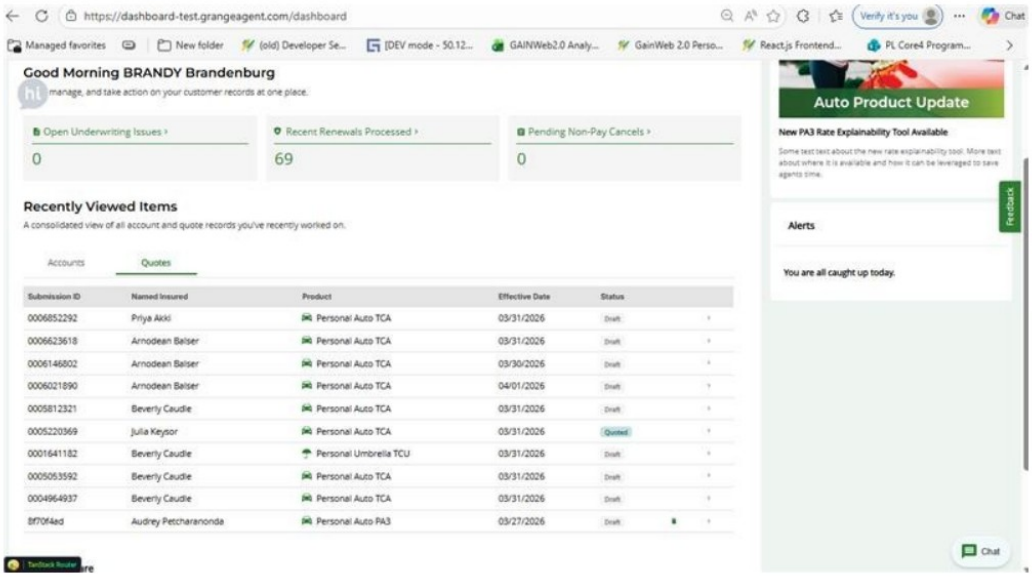


Figure: supplied portal screen – KYP presentation page 20.

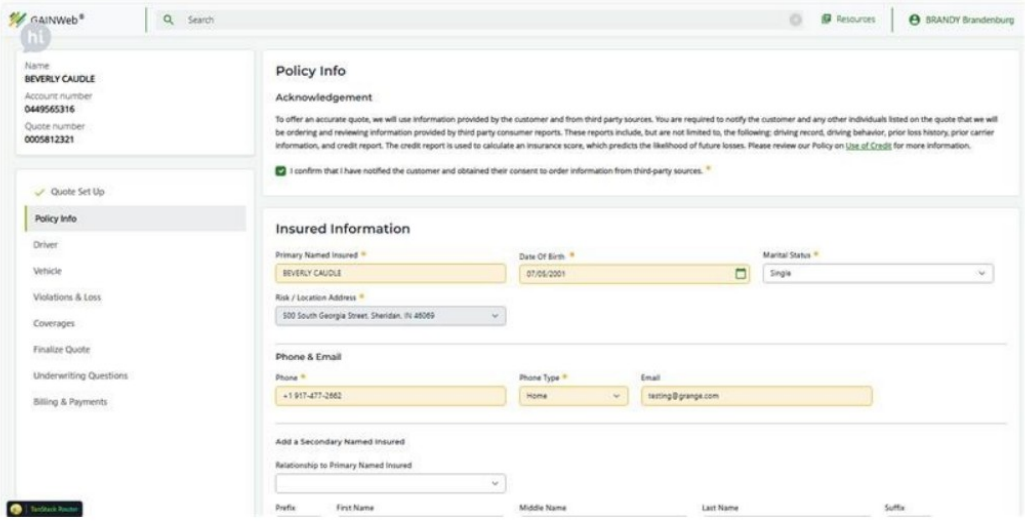


Figure: supplied portal screen – KYP presentation page 21.

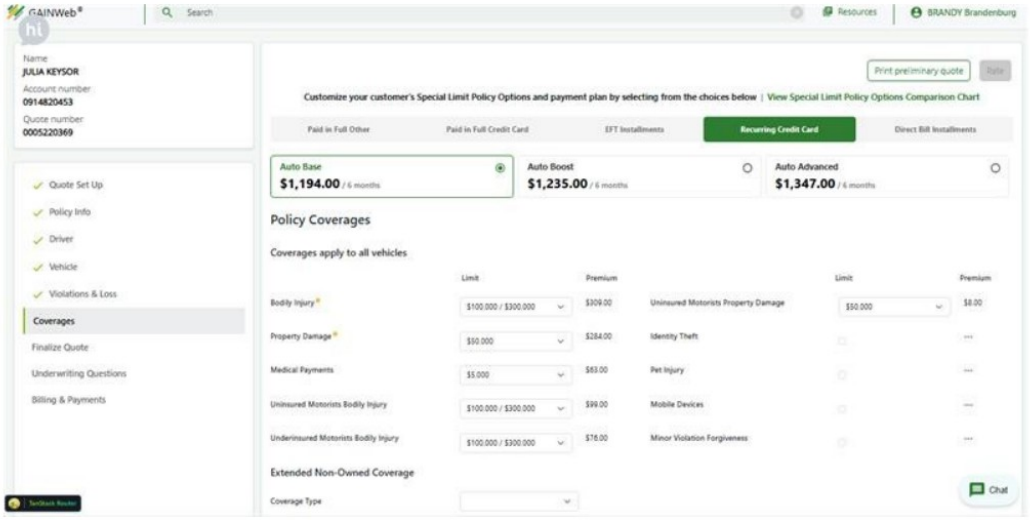


Figure: supplied portal screen – KYP presentation page 22.

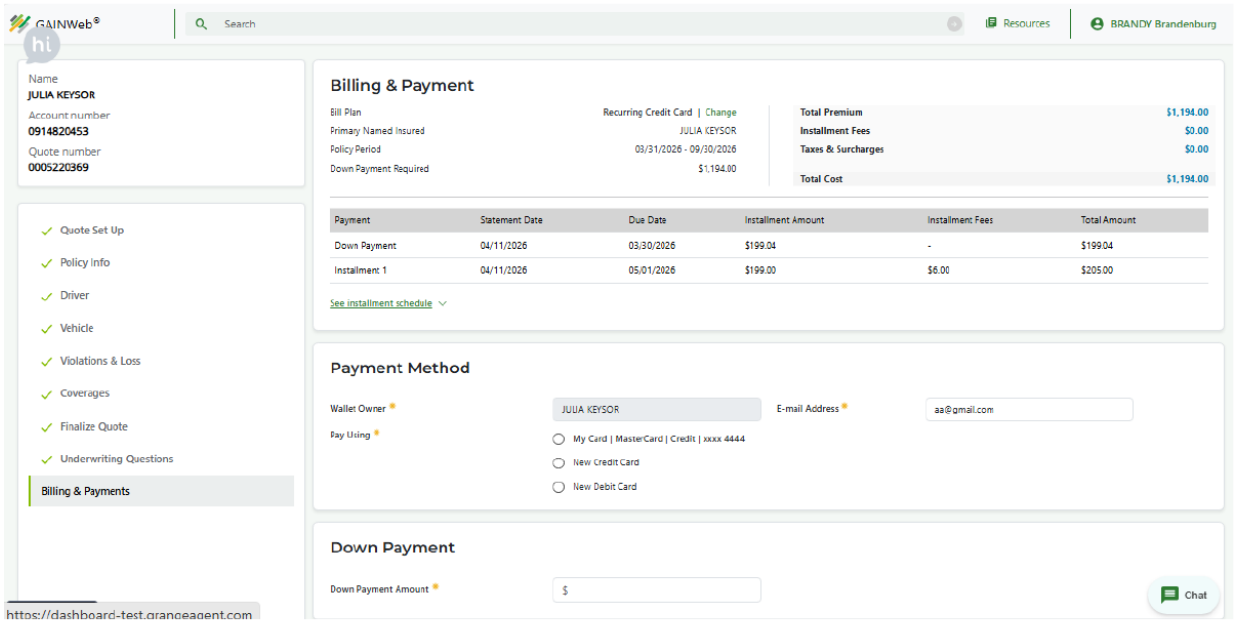


Figure: supplied portal screen – KYP presentation page 23.

4. Personas & Access

The technical deck identifies Agents and Internal Users as the primary personas. It states that there is no role hierarchy within the agency and that most functionality is shared, with specific restrictions for Internal Users.

Persona	Access / behavior from source
Agent	Full agent-facing portal capabilities within the defined scope.
Internal User	Same functionality except metrics are not shown and Account/Transactions menu items are disabled.

Authentication and token management are shown through Agency Access Management. The NFRs require authorized-user validation through Agency Access Management.

5. Target Solution Architecture

The supplied architecture diagram shows the portal behind F5 Load Balancer and connected to Agency Access Management, an ESB, AWS API Gateway/Lambda/DynamoDB components and multiple policy/legacy/external systems. A second architecture view adds CloudFront, WAF, S3 application bundles/logs, Secrets Manager, an authorizer Lambda and on-premises F5/ESB resources.

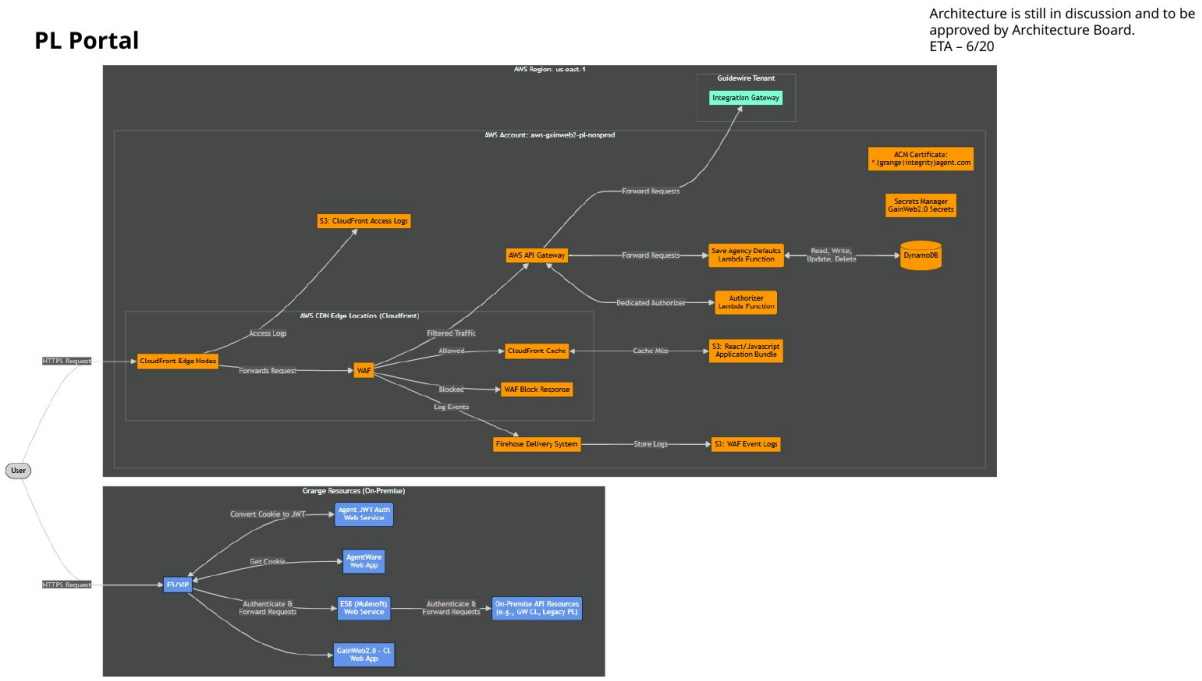


Figure: target-state architecture view from the supplied PL Core4 architecture deck (page 8).

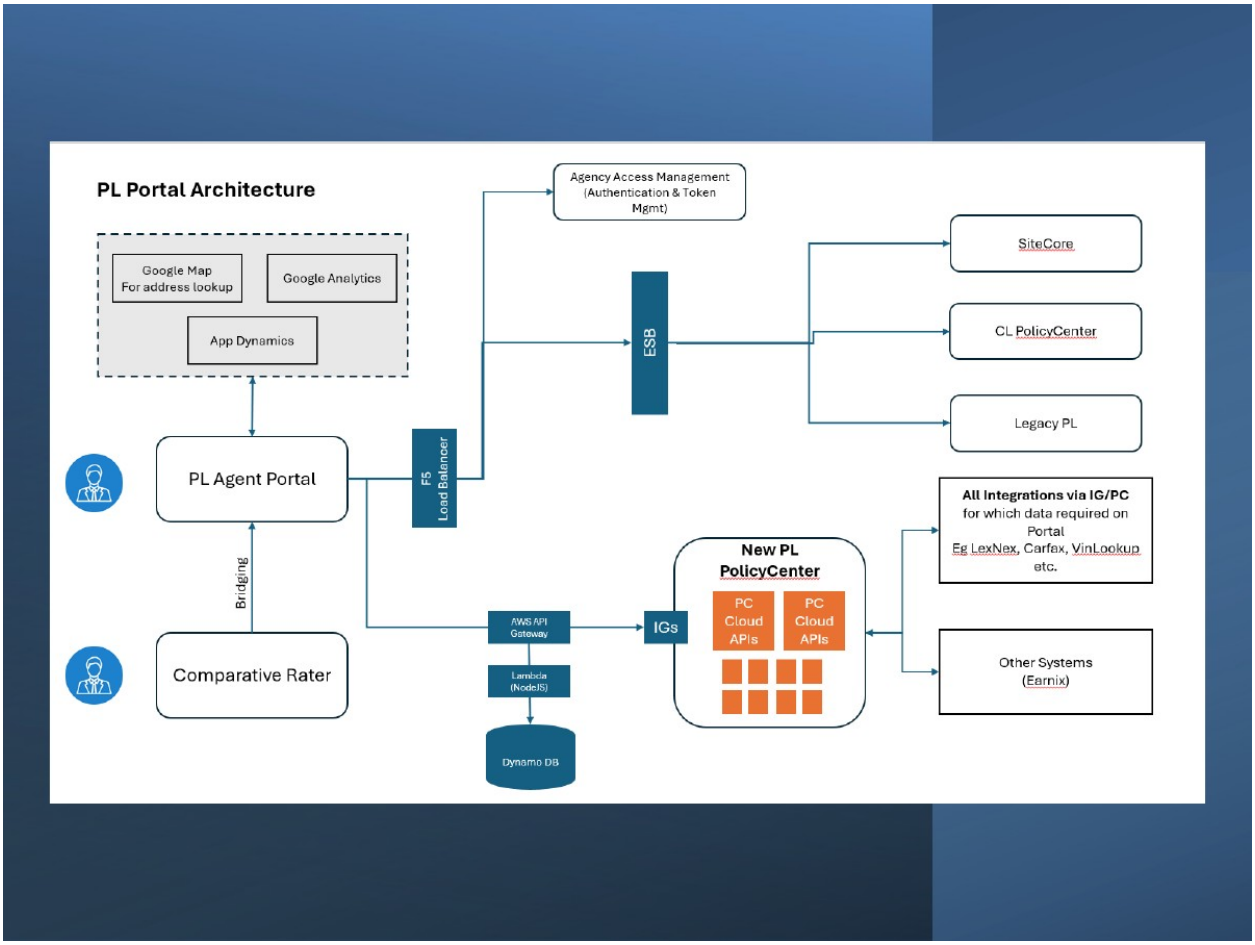


Figure: logical portal architecture from the KYP material (page 3).

6. Architecture Layers

Layer	Components / responsibilities evidenced in source
Experience / UI	React 19, TypeScript, React-Bootstrap, responsive UI, reusable components, global search, dashboard and quote journeys.
State / client data	Zustand and TanStack React Query; quote execution notes state that data is held in state management and PolicyCenter is called at logical step boundaries, with exact timing to be defined during development.
Routing / HTTP	TanStack Router and Axios.
Edge / delivery	F5 Load Balancer; CloudFront; WAF; CloudFront cache; S3 React/JavaScript bundle.
API / integration	AWS API Gateway, Lambda (NodeJS), ESB / Integration Gateway and PolicyCenter cloud APIs.
Security	Agency Access Management; token management; dedicated authorizer Lambda; ACM certificate; Secrets Manager.
Systems of record / enterprise	New PL PolicyCenter, CL PolicyCenter, Legacy PL and other systems such as Earnix.
External / enrichment	LexisNexis, Carfax, VIN Lookup and other integrations identified in quote flows; Google Maps for address lookup.
Observability	AppDynamics; ELK / centralized logging; CloudFront and WAF access/event logging.
Analytics	Google Analytics with PII masking through the private-info CSS attribute; the project workbook also references analytics/experimentation tools.

7. Technology Stack

Technology	Version / source detail	Purpose
React	19.0	Declarative UI library
TypeScript	5.8.3	Typed JavaScript development
Zustand	5.0.5	Client state management
React-Bootstrap	2.10.10	Reusable Bootstrap UI components
TanStack Router	1.120.5	Type-safe routing
TanStack React Query	5.77.1	Server-state fetching/caching/synchronization
Axios	1.9.0	HTTP client
ESLint	9.27.0	Linting/code quality
Webpack	5.99.9	Module bundling/code splitting
Babel	7.27.2	JavaScript compilation
React Testing Library	Compatible	UI testing
Jest	29.x	Unit testing

The project-information workbook additionally describes a React.js microfrontend architecture, shared libraries, React Hook Form, Zustand, Playwright-based E2E automation and CI/CD. Where these statements differ in emphasis from the portal technical deck, both have been retained as source-derived project information.

8. Integration Architecture

The portal is integration-heavy. The supplied material identifies PolicyCenter, CL PolicyCenter, Legacy PL, ESB/Integration Gateway, Agency Access Management and several external data providers. The architecture objective is to keep integration failures from taking down the UI and to provide meaningful user-facing messages when dependent services are unavailable.

Integration category	Examples from source	Architectural consideration
Authentication	Agency Access Management; token management	Validate authorized users and manage tokens.
Policy / quote	New PL PolicyCenter; CL PolicyCenter; Legacy PL	Portal-to-system integration and consistent user flows.
Enterprise integration	ESB / Integration Gateway	Central integration path for required portal data.
External enrichment	LexisNexis, Carfax, VIN Lookup, Equifax, CLUE, FraudScore, etc.	Resilience, timeout/error handling and meaningful fallback messaging.
Address / analytics	Google Maps; Google Analytics	PII protection and graceful degradation.

Integration design principles

- Isolate external failures from the portal experience.
- Use retry/fallback mechanisms where appropriate.
- Apply API Gateway throttling where applicable.
- Provide meaningful messages when dependent systems are unavailable.
- Maintain secure token-based authentication.
- Capture logs and telemetry for troubleshooting.

9. Quote Journey Architecture

9.1 Auto Quote

- Dashboard → Start Quote → customer/address → product selection → policy info → driver → vehicle → coverages → request quote → display quote → underwriting decision → submission/purchase/issue → payment/e-sign → policy/document link

The source identifies integrations such as VIN Lookup, Carfax, InsureScore, Equifax, MVR, CLUE, telematics, fraud/risk services, coverage defaults and policy creation.

9.2 Home Quote

- Dashboard → Start Quote → customer/address → policy info and prefill → risk address → HO3/HO4/HO6 → questions → coverages → request quote → display quote → underwriting decision → submission/purchase/issue → payment/e-sign → policy/document link

The source identifies enrichment/integration services including InsureScore, ProtectionClass, CLUE, FraudScore, RiskAnalyzer, 360Value, Equifax, A-Plus, Zesty.ai and BuildFax.

9.3 Umbrella Quote

- Dashboard → Start Quote → customer/address → available policies → additional details → coverages → request quote → display quote → eligibility/underwriting → submission/issue → payment/e-sign → policy/document link

10. Non-Functional Requirements & Quality Attributes

Quality attribute	Source requirement / architecture response
Performance	Response time within 2 seconds, subject to alignment with Policy response-time agreement.
Availability / resilience	External integrations should not bring down the app; meaningful messages required when systems are down.
Security	Authorized access validated with Agency Access Management; token-based authentication.
Accessibility	WCAG 2.1 AA; fonts, font sizes, color scheme and help text are explicitly called out.
Responsive UI	Support different breakpoints / devices.
Testing	Minimum 80% unit test coverage to pass Sonar quality gate.
Observability	ELK logging and AppDynamics monitoring.
Analytics privacy	Mask PII sent to Google Analytics using the private-info CSS attribute.

11. Engineering & Delivery Governance

The source material describes Agile/Scrum, Git-based development, CI/CD, Sonar quality gates, internal code review and reusable components. The supplied pipeline screenshot shows stages including prepare, lint, audit, test, build and deploy.

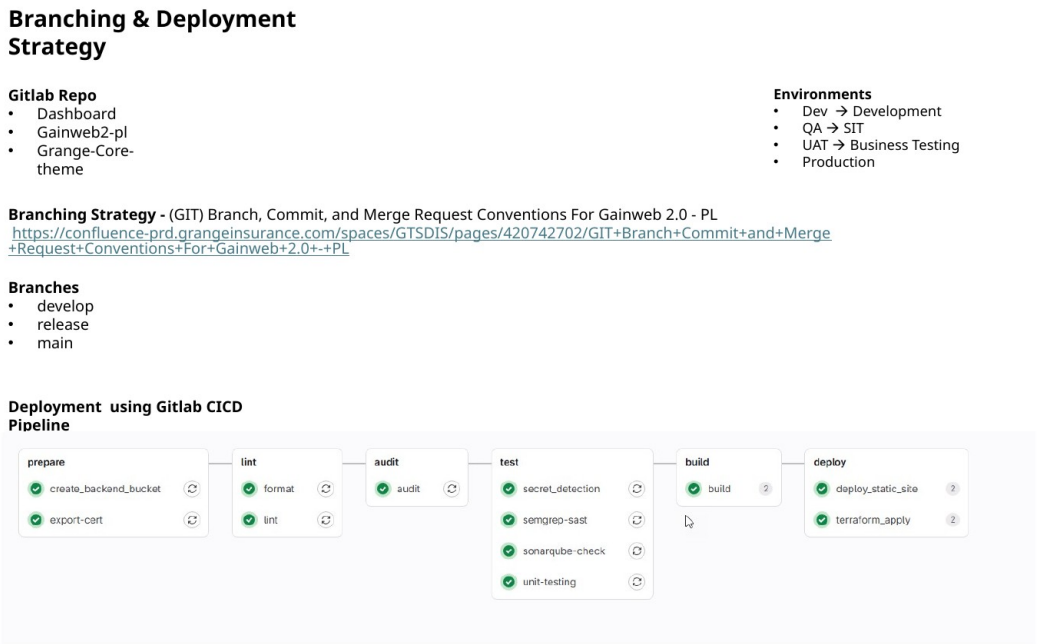


Figure: supplied GitLab CI/CD pipeline and branching/deployment view (technical architecture deck page 18).

Area	Architecture governance responsibility
Architecture	Maintain target architecture and record architecture decisions.
Design	Review component, API, integration and data design.
Code quality	Ensure coding standards, SOLID practices and Sonar quality gates are followed.
Testing	Ensure unit/integration/E2E strategy supports the NFRs.
Security	Validate authentication, token handling, PII controls and secure coding.
Delivery	Review CI/CD, deployment, rollback and production-readiness approach.
Technical debt	Identify and prioritize architectural/design debt.

12. Architecture Risks & Mitigations

Risk / challenge	Source evidence	Mitigation / architecture response
Multi-system integration	PolicyCenter, CL, Legacy PL and external systems	Integration abstraction, retry/fallback, monitoring and meaningful errors.
Real-time data synchronization	Portal must interact with multiple enterprise systems	Define state ownership and logical integration boundaries; avoid unnecessary client-side duplication.
Legacy dependency	Legacy PL is part of the portal scope	Isolate legacy integration behind integration services / ESB where applicable.
Performance bottlenecks	<2 sec target	Caching, optimized API calls, responsive UI, performance testing and monitoring.
Authentication/security	Agency Access Management and token handling	Centralized authentication/token management and authorization.
Third-party dependency failures	Google Map/SiteCore and external data providers	Graceful degradation, fallback/error messaging and telemetry.
Accessibility	WCAG 2.1 AA	Accessibility review during design, development and testing.
Quality gate failure	80% minimum unit coverage	Automated tests, Sonar quality gate and CI enforcement.

13. Solution Architect Responsibilities – Grange Context

Lifecycle stage	SA responsibilities for this portal
Discovery	Understand agent workflows, business scope, personas and current enterprise systems.
Requirements	Drive technical requirements and NFRs for performance, security, accessibility, resilience and observability.
Architecture	Own the portal logical/target architecture and integration boundaries.
Technology	Govern React/TypeScript/UI/state-management/tooling choices and shared component strategy.
Integration	Coordinate PolicyCenter, CL, Legacy PL, ESB/API Gateway and external service integrations.
Security	Work with Agency Access Management and security stakeholders on authentication, authorization, tokens and PII.
Development	Guide implementation, design reviews, code-quality practices and reusable components.
Testing	Ensure NFR validation, unit coverage, integration testing and E2E automation.
Deployment	Review GitLab CI/CD, environment promotion and production readiness.
Production	Support incident/RCA analysis, performance monitoring, technical debt and architecture evolution.

14. Architecture Decision Records – Recommended Set

ADR	Decision topic	Why it matters
ADR-001	Portal integration pattern	Define portal → API Gateway/ESB/PolicyCenter boundaries.
ADR-002	Client state vs server state	Clarify Zustand vs TanStack Query ownership.
ADR-003	Quote-step persistence strategy	Define when PolicyCenter is called and what remains in client state.
ADR-004	External integration resilience	Define timeout, retry, fallback and user-message behavior.
ADR-005	Authentication/token flow	Define Agency Access Management and authorizer interaction.
ADR-006	Analytics and PII masking	Define Google Analytics tagging and privacy controls.
ADR-007	Performance strategy	Define caching, payload, rendering and API performance targets.
ADR-008	Accessibility strategy	Define WCAG 2.1 AA implementation and validation.
ADR-009	CI/CD quality gates	Define lint, audit, unit test, Sonar and deployment gates.

15. Production Readiness Checklist

- Architecture and key ADRs approved.
- All critical integrations validated in the target environment.
- Authentication and authorization flows validated.
- Performance target and Policy response-time dependency validated.
- External-service failure behavior tested.
- WCAG 2.1 AA checks completed.
- Google Analytics PII masking validated.
- Minimum 80% unit coverage and Sonar quality gate passed.
- Centralized logging and AppDynamics monitoring operational.
- CI/CD deployment path validated.
- Rollback/cutover approach documented.
- Production support and incident ownership agreed.

16. Open Architecture Topics / Items to Confirm

The following items are explicitly identified as discussion, dependency or to-be-finalized topics in the supplied material. They should be closed through architecture decisions rather than assumed.

- Architecture Board approval status.
- Final approach for wrapper APIs versus direct portal integration with the three major systems.
- Exact timing of PolicyCenter calls at each logical quote step.
- Final response-time agreement with Policy.
- Scope and behavior of items noted as still under discussion.
- Final environment-specific infrastructure configuration.
- Final production observability and alert thresholds.

17. Reference to Supplied UI Screens

The supplied screens show a coherent end-to-end agent journey:

Screen	Architectural significance
Dashboard	Aggregates agency-level work items and recently viewed data.
Select Product	Entry point for Auto, Home and Umbrella products.
Account Details	Aggregates policy, billing and claims-related account information.
Quote List	Provides filtered quote visibility and status.
Policy Info	Captures insured information and quote setup details.
Coverages	Supports coverage selection and premium/limit presentation.
Billing & Payment	Completes payment-plan and payment-method processing.

The UI evidence aligns with the source-defined portal scope and quote workflows and should be treated as the experience layer over the integration architecture described above.

18. Architecture Summary

The Grange Personal Lines Agent Portal 2.0 can be viewed as a modern React-based experience layer over a heterogeneous insurance ecosystem. The central architectural challenge is not the UI alone; it is the controlled orchestration of authentication, portal state, PolicyCenter/legacy integrations, external enrichment services, resilience, observability, accessibility and CI/CD quality controls.

For the Solution Architect, the key responsibility is to keep the business journey and the technical landscape aligned: every screen should map to a defined capability, API/integration boundary, security control, NFR, operational behavior and production-support model.

Source-grounded note: this document intentionally does not introduce unsupported infrastructure, API contracts, database schemas, SLAs or production topology beyond what is visible in the supplied files. Those items should be added as the authoritative project artifacts become available.

Prepared as an architecture working document from the supplied Grange project materials.